NES AI FINANCE & INSURANCE INTELLIGENCE PLATFORM

Market Data Source Specification

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Build approach: Cline-assisted development • Local-first Windows application

Scope: Personal research, paper trading and insurance document assistance

Provider evaluation, adapter contract and freshness rules

1. Purpose

This specification defines what must be evaluated before connecting an external market data provider. No provider should be treated as available or licensed until its current terms and access have been verified.

2. Provider selection checklist

- NSE/BSE instrument coverage and supported segments.
- Real-time versus delayed data, timestamps and market-session semantics.
- Historical intraday/daily data availability and retention.
- Commercial/personal-use licence terms and redistribution restrictions.
- API rate limits, WebSocket support and reconnection behaviour.
- Corporate actions, adjusted data and instrument identifiers.
- Authentication flow, token expiry and secret storage.
- Cost, reliability, support and provider status endpoint.

3. Adapter contract

Operation	Required behaviour
get_market_status	Return status, source and timestamp where supported.
search_instruments	Normalise symbols and exchange identifiers; handle ambiguity.
get_quote	Return value, currency, source, timestamp and data status.
get_history	Return bars with timezone, interval, source and missing-data metadata.
subscribe_updates	Optional; reconnect safely and distinguish delayed/unavailable feeds.
health_check	Check provider connectivity, not whether the market itself is open.

4. Freshness and failure states

- Represent LIVE, DELAYED, STALE, CLOSED, DISCONNECTED, PARTIAL and DEMO as distinct states where appropriate.
- A closed market does not mean the data connection is broken.
- Never replace missing timestamps with the current time.
- Show the last successful update and the source of the value.
- On provider failure, preserve last-known data only if visibly labelled stale.

5. Trading and licence boundaries

- Do not route orders through the market-data adapter.
- Broker connectivity, if ever considered, requires a separate design and review.
- Confirm the provider's terms for local storage, historical downloads and any future customer-facing use.
- This specification does not select or endorse a particular provider.

6. Prototype rule

Until a provider is selected and tested, the UI must use DEMO_DATA labels and must never show a “Live” badge.